

Chapter 2: Probability

Sample space: Set of all possible outcomes.

Event: Subset of sample space

Multiplication rule:

throwing a die 2 times

$$\left. \begin{array}{l} 1. \text{ step } 6 \text{ diff. sit. } \Rightarrow n_1 \\ 2. \text{ " } 6 \text{ " " } \Rightarrow n_2 \end{array} \right\} n_1 \cdot n_2 = 36$$

Permutation: $\frac{n!}{(n-r)!}$

Combination: $\binom{n}{r} = \frac{n!}{(n-r)! \cdot r!}$

Probability of an event:

$$0 \leq P(A) \leq 1 \quad P(\emptyset) = 0 \quad P(S) = 1$$

Equally likely outcomes: If an experiment has N equally likely outcomes, and event A occurs in n ways, then $P(A) = n/N$.

Additive Rules:

General: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Mutually exclusive: $A \cap B = \emptyset$; $P(A \cup B) = P(A) + P(B)$

Independence: $P(A \cap B) = P(A) \cdot P(B)$

Rule of Complements: $P(A) + P(A') = 1$

Conditional Probability: $P(B|A) = \frac{P(A \cap B)}{P(A)}$ $P(A) > 0$

Product Rule: $P(A \cap B) = P(A) \cdot P(B|A)$

Baye's Rule:

$$\begin{array}{ccc} \frac{M_1}{\frac{0.3}{0.1}} & \frac{M_2}{\frac{0.3}{0.2}} & \frac{M_3}{\frac{0.4}{0.3}} \\ & & \frac{0.2}{0.1+0.2+0.3} \end{array}$$

Answers question: "I originally thought the probability of A was this. Now that B has happened, what is the new probability of A?"

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Initial belief about A

The probability of A after you see B.

Total probability of B happening

The probability of seeing B, assuming A is true.

Law of Total Probability:

Finding probability of an event when you don't know the answer directly, but you do know its probability under different conditions.

$$P(A) = \sum_{i=1}^k P(B_i | A) = \sum_{i=1}^k P(B_i) \cdot P(A | B_i)$$

Chapter 3: Random Variable and Prob. Dist.

Numerical outcome of random experiment.

Discrete random variable: Has a finite or countable.

Continuous random variable: Has an uncountable number of outcomes.

Probability Distributions:

Probability Mass Function (PMF):

For discrete:

$$f(x) \geq 0$$

$$\sum_{\text{all } x} f(x) = 1$$

For continuous:

$$f(x) \geq 0$$

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

Joint Probability Dist.

Two or more random var.

Discrete (Joint PMF):

$$f(x, y) = P(X=x, Y=y)$$

The sum of all entries in the table must be 1

Continuous (Joint PDF): 3D surface where the total volume under must be 1

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

Cumulative Distribution Function (CDF)

Def: $F(x) = P(X \leq x)$

For discrete: $F(x) = \sum_{t \leq x} f(t)$

For continuous: $F(x) = \int_{-\infty}^x f(t) dt$

A key use of CDF is finding prob. of range.

$$P(a < X \leq b) = F(b) - F(a)$$

Marginal Distribution: Prob. dist. of just one var., which you get from joint dist.

Discrete: you sum the columns of the table

$$g(x) = P(X=x) = \sum_{\text{all } y} f(x, y)$$

Continuous: Integrate over y.

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

Conditional dist: Joint / Marginal

$$\text{Discrete: } f(x|y) = P(X=x | Y=y) = \frac{f(x, y)}{h(y)}$$

Statistical Independence:

$$f(x, y) = g(x) \cdot h(y)$$

Chapter 4: Expected value (mean) and Variance

Mean:

- For discrete var:

$$\mu = E[X] = \sum_{\text{all } x} x \cdot f(x)$$

- For continuous var:

$$\mu = E[X] = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

Variance (σ^2)

$$\sigma^2 = \text{Var}(X) = E[X^2] - (E[X])^2$$

Discrete: $E[X^2] = \sum_{\text{all } x} x^2 \cdot f(x)$

Continuous: $E[X^2] = \int_{-\infty}^{\infty} x^2 \cdot f(x) dx$

Expectation of a Function, $E[g(x)]$

$$Y = g(x)$$

Yerine key ana gire islem yap.

$$E[g(x)] = \int_{-\infty}^{\infty} g(x) \cdot f(x) dx$$

Chapter 5: (Discrete Dist.)

1. Binomial Distribution:

Used when you have a fixed number (n) of independent trials, and each trial has only two outcomes: "success" (with prob. p) or "failure" (with prob. $q = 1-p$)

- Finds prob. of getting exactly x successes in those n trials.

$$b(x; n, p) = \binom{n}{x} \cdot p^x \cdot q^{n-x} \quad \left\{ \begin{array}{l} \mu = n \cdot p \\ \sigma^2 = n \cdot p \cdot q \end{array} \right.$$

2. Poisson Distribution

used when you are counting "successes" that occur in a continuous interval.

$$p(x; \lambda t) = \frac{e^{-\lambda t} \cdot (\lambda t)^x}{x!}$$

the average number of events in that specific interval.

$$\mu = \lambda t \quad \left\{ \begin{array}{l} \sigma^2 = \lambda t \end{array} \right.$$

*Poisson approx. to the Binomial

You can use poisson dist. to approx. a Binomial problem when;

- 1) n is very large
- 2) p is very small

$$\mu = np = \lambda t$$

3. Multinomial Distr: More than two possible outcomes.

$$f(x_1, x_2, \dots, x_k; p_1, p_2, \dots, p_k, n) = \binom{n}{x_1, x_2, \dots, x_k} \cdot p_1^{x_1} \cdot p_2^{x_2} \cdot \dots \cdot p_k^{x_k}$$

4. Hypergeometric Distr: (two types "defective and good")

Use this when you are sampling without replacement. The trials are not independent.

N : Destedeki bütün kartlar

K : Başarı sayısı

$N-K$: Başarısız

n : elinen kart sayısı

x : istenilen başarı sayısı

$$h(x; N, n, k) = \frac{\binom{k}{x} \binom{N-k}{n-x}}{\binom{N}{n}}$$

5. Multivariate Hypergeometric (three or more types)

Sample without replacement from a group :

$$P(x_1, x_2, \dots, x_k; a_1, a_2, \dots, a_k, N, n) = \frac{\binom{a_1}{x_1} \cdot \binom{a_2}{x_2} \cdots \binom{a_k}{x_k}}{\binom{N}{n}}$$

Number of success you want for each type.

The number of items of each type

total numbers of items

total sample size

6. Negative Binomial & Geometric Distribution

- **Geometric Story:** Instead of fixing n , you count the number of trials (x) needed to get your first success.
 - **Formula:** $g(x; p) = pq^{x-1}$ (This is one form, $x = 1, 2, 3, \dots$).
- **Negative Binomial Story:** A general version of the Geometric. You count the number of trials (x) needed to get your k -th success.
 - **Formula (from cheat sheet):** $b^*(x; k, p) = \binom{x-1}{k-1} p^k q^{x-k}$.
 - Note: The Geometric is just a Negative Binomial where $k = 1$.

Chapter 6:

Uniform Distribution:

Belli bir aralıktaki tüm sonuçlar eşit derece olası $\Rightarrow$ uniform

A: min değer

B: max değer

$$f(x) = \frac{1}{B-A} \Rightarrow \text{Density func}$$

$$\mu = \frac{A+B}{2}$$

$$\sigma^2 = \frac{(B-A)^2}{12}$$

How to Use It:

- The PDF is a flat horizontal line (a rectangle) between A and B .
- Finding probabilities is as simple as finding the area of a rectangle: $\text{Area} = \text{width} \times \text{height}$.
- Example:** To find $P(c < X < d)$, where $[c, d]$ is inside $[A, B]$:
 - The width is $(d - c)$.
 - The height is $\frac{1}{B-A}$.
 - So, $P(c < X < d) = (d - c) \cdot \frac{1}{B-A}$.

Normal Distribution: This is the "bell curve".

$$n(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Mean: This is the center of the bell. (μ)

Variance: σ^2 . (How "spread out" the bell is)

How to use;

1. Standardize (Find z-score):

$$z = \frac{X - \mu}{\sigma}$$

2. Using z-table

• find $P(X < b)$

- calc $z_b = \frac{b - \mu}{\sigma}$

- look up $P(Z < z_b)$ in the table.

• find $P(a < X < b)$

- calc $z_a = \frac{a - \mu}{\sigma}$

- calc $z_b = \frac{b - \mu}{\sigma}$

- Find $P(z_a < Z < z_b) = P(Z < z_b) - P(Z < z_a)$

To go back:

$$X = \mu + z\sigma$$

Gamma Distribution

$$\Gamma(\alpha) = (\alpha - 1)!$$

α : beklenen olay sayısı

β : iki olay arası ortalama geçen süre

$$f(x; \alpha, \beta) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} \cdot x^{\alpha-1} \cdot e^{-x/\beta} & , x > 0 \\ 0 & , \text{elsewhere} \end{cases}$$

$$\mu = \alpha \cdot \beta$$

$$\sigma = \alpha \cdot \beta^2$$

Exponential:

İlk durumun gerçekleşmesi için geçen süre.

$$\alpha = 1$$

• If you set $\alpha = 1$ in gamma.

$$f(x; \beta) = \frac{1}{\beta} \cdot e^{-x/\beta}$$

Mean: $\mu = \beta$

Variance: $\sigma^2 = \beta^2$

Relationship to Poisson:

$$\beta = 1 / \lambda$$